ovid CS how-to guide

One task per section: when to use it, the steps, what changes and how to check it. Page names are those of the Standard view: the stages are on the top bar, the pages of the stage shown are listed on the left, and those under **More pages** are folded at the bottom of that list; **Go to** (⌘K or Ctrl K) finds any page. Numbers in brackets point to the markers in the figure. For what CS does to the data and why, see the CS reference guide.

Contents

Bring in: 1. Add a new download; 2. Send a file to another station or leave it out; 3. Name the columns of CR10X files

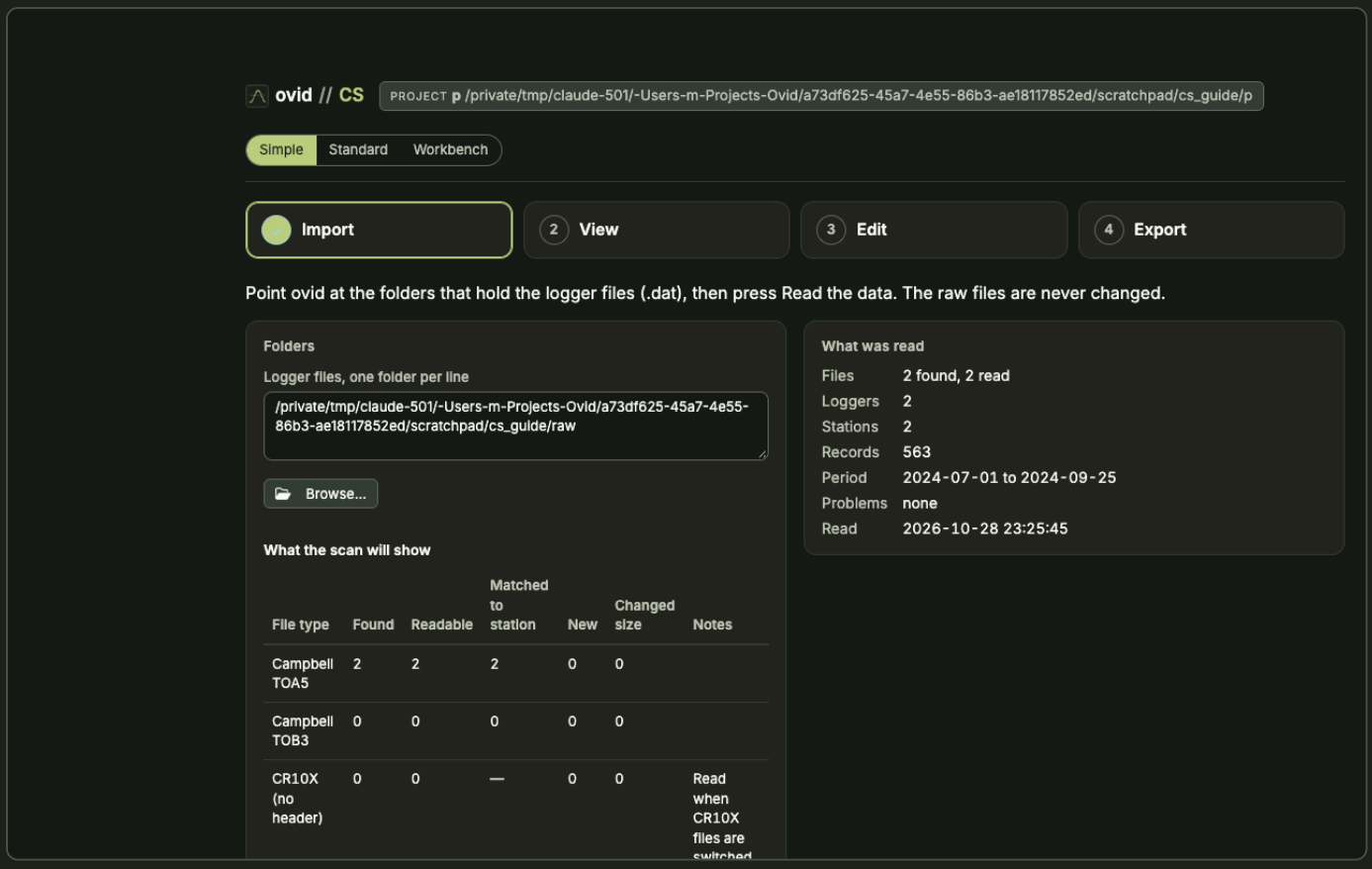
Organize: 4. Give a logger a station; 5. Give a raw column a common name; 6. Combine stations into one record

Correct: 7. Fix a logger clock; 8. Set the time zone; 9. Remove a bad stretch; 10. Add a calibration; 11. Change a quality limit; 12. Set up a weighing gauge; 13. Get evapotranspiration and albedo

Check and report: 14. Find out why a value was removed; 15. Find failing sensors; 16. Export data and reports; 17. Back up or share the set-up; 18. Report a problem

The three views

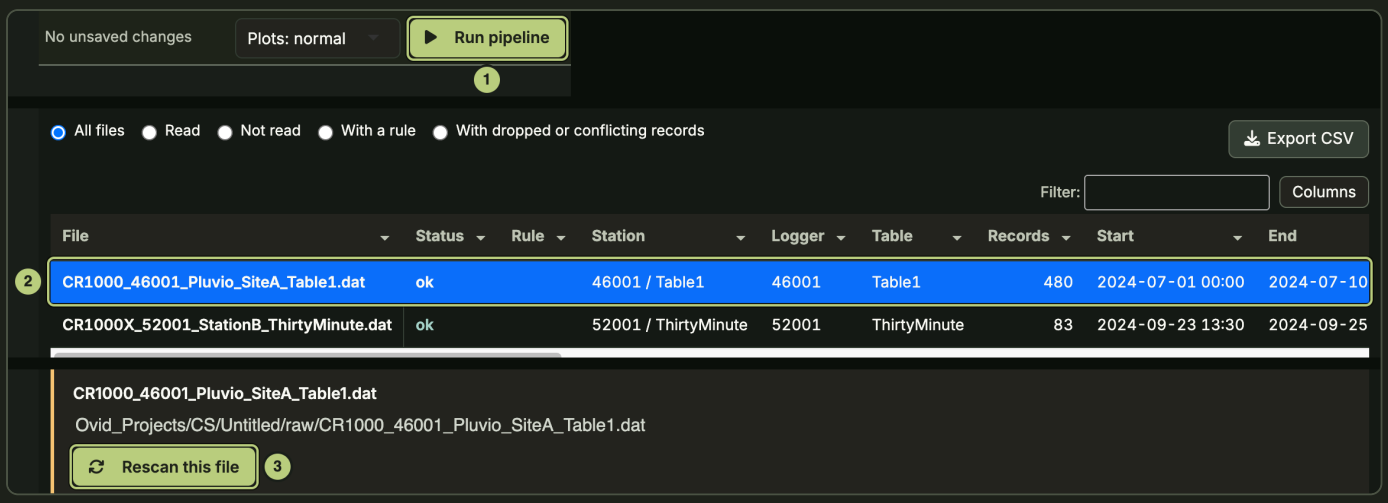
The same project, data and edits in three views, switched at the top (**Simple**, **Standard**, **Workbench**) or with ?ui=simple, ?ui=standard or ?ui=workbench in the address. A new install opens in Simple.



- **Simple**: a field check in four steps (Import, View, Edit, Export). Edit renames stations, places them and moves loggers between them; it changes no data value.
- **Standard**: the set-up flow these tasks use: bring in files, organize loggers and stations, assess the records, decide what the run found, report.
- **Workbench**: every change to the data. Removing values, quality limits, corrections over a period and clock shifts made by hand are edited here; the Standard view shows them and links to it.

1. Add a new download

When: after a field visit, with new .dat files.



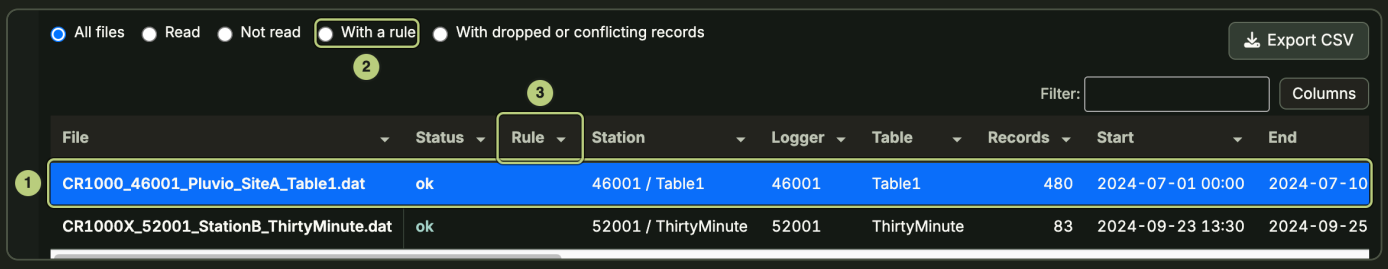
1. Copy the files into a raw folder of the project. Keep the overlapping old ones.
2. Press **Run** (1).

Result: only logger tables with new or changed files are read again; the rest come from cache/. Overlapping records are merged, the newest download's values winning.

Check: 2 Bring in files lists each new file and where its records went (2). To read a file again, select it and press **Rescan this file** (3).

2. Send a file to another station or leave it out

When: a file belongs to another logger, is a bad copy, or should not be used.



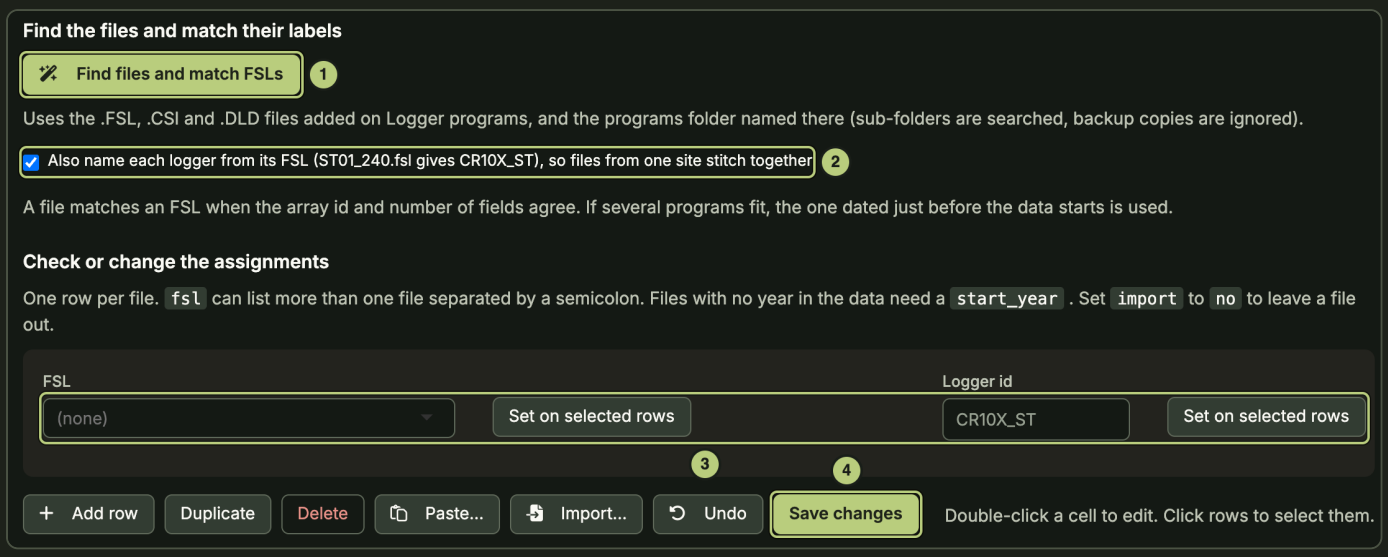
1. **2 Bring in files:** select the file (1).
2. Below its details, give a reason and your initials, then **Send to this station**, **Leave out of the next run**, or drag across its plot and **Remove these records from this file**.
3. Press **Run**. The rule is saved at once.

Result: a file rule in the set-up tables, applied on the next run. The raw file is not changed.

Check: With a rule (2) lists the file; the **Rule** column (3) shows it. The **Audit trail** says why.

3. Name the columns of CR10X files

When: CR10X mixed-array files show columns without names.



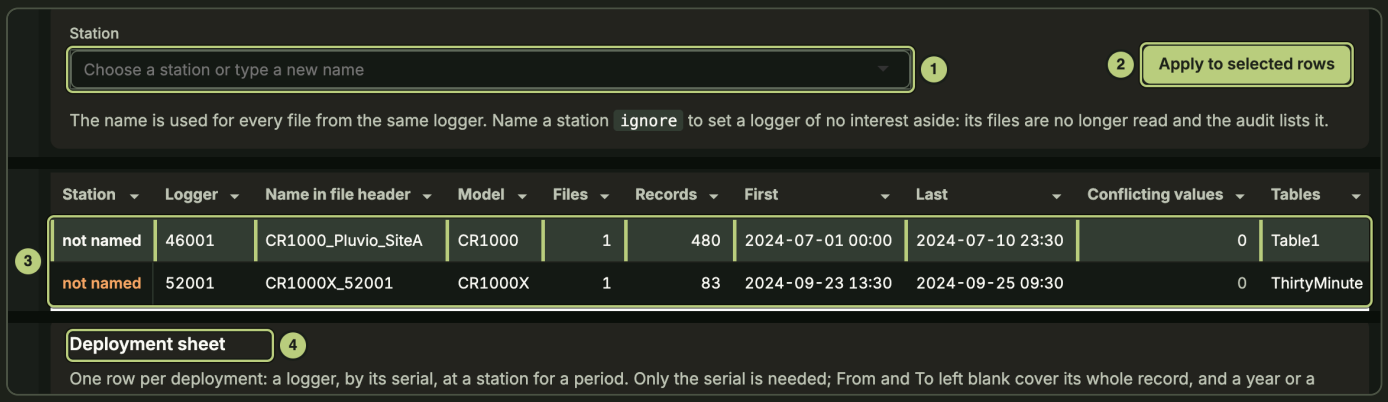
1. Add the .FSL files on **Logger programs** (More pages).
2. **CR10X files** (More pages): press **Find files and match FSLs** (1). Tick **Also name each logger from its FSL** (2) so files from one site stitch together.
3. Check each file has an FSL and a logger id; set missing ones with **Set on selected rows** (3), then **Save changes** (4).
4. Press **Run**.

Result: each array gets time stamps and column names from its FSL.

Check: Loggers and files (More pages) lists the logger with its columns named.

4. Give a logger a station

When: a new logger, a logger moved, or two loggers make up one station.



1. **3 Loggers and stations:** select the logger rows (3).
2. Pick the **Station** (1) and press **Apply to selected rows** (2).
3. A logger that moved: one row per station and period in the **Deployment sheet** (4).
4. To set a logger aside, name its station **ignore**: its files are not read. Press **Run**.

Result: records go to stations by logger, header name, table and date.

Check: Logger history (More pages), tab **Station lineage**.

5. Give a raw column a common name

When: a column keeps its raw name (for example a new sensor or a renamed program variable).



1. **Column names** (More pages): pick the column under **Raw columns with no alias yet** (1) and the **Common parameter** (2).
2. Press **Add alias** (3), then **Run**.

Result: the raw column is renamed to the common parameter (AirTC_Avg becomes air_temp) in every table that has it.

Check: the **All columns** tab (4); output/audit/column_dictionary.csv.

6. Combine stations into one record

When: one parameter needs a single record from several stations, for example a sensor that moved.



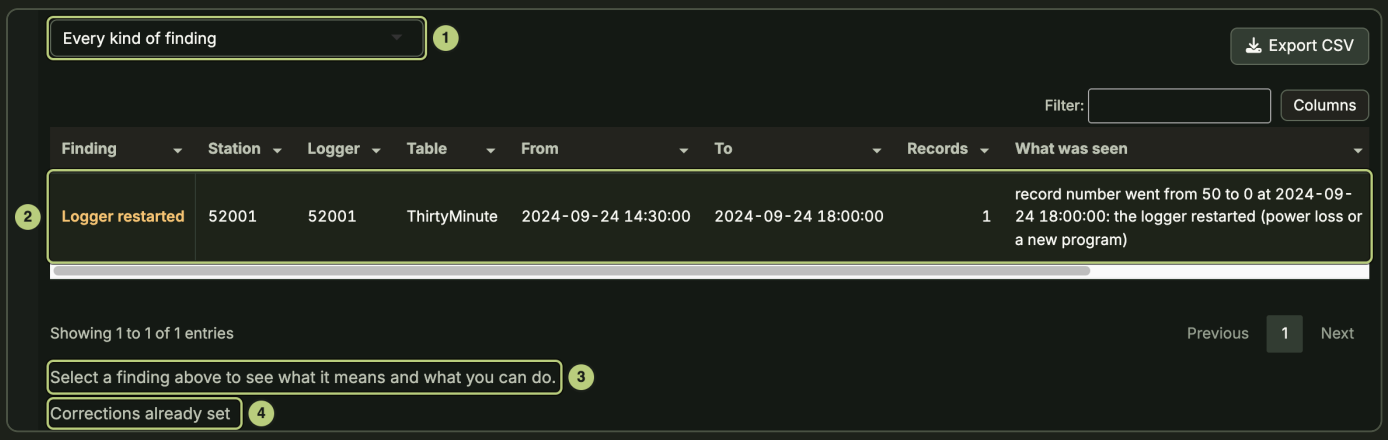
1. **Combined records:** name the **Combined record**, pick its **Parameter** and **Time step** (1).
2. **Add source** (2) for each station and table. Limit each with **From** and **To**; give an **Offset** if needed.
3. Order the sources with **Move up** and **Move down** (3): where two have a value, the higher one wins.
4. **Save** (4).

Result: the record is rebuilt on every run into output/composites/. Each value keeps the station it came from.

Check: Export CSV on the page.

7. Fix a logger clock

When: Clocks lists a finding (a power loss, a reset clock, overlapping files that disagree in time).



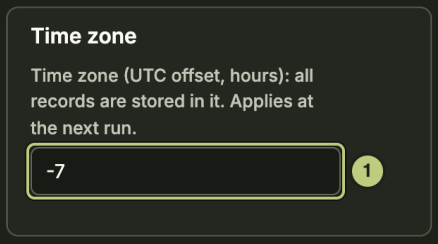
1. **Clocks:** pick the kind (1) and select the finding (2). Read what it means below (3).
2. A suggested shift: accept it here or in **Needs you**. Any other correction is made in the Workbench: **Move the timestamps** (by minutes, or the real time of the first record) or **Remove the records**, with a reason, then **Add this correction**.
3. Press **Run**.

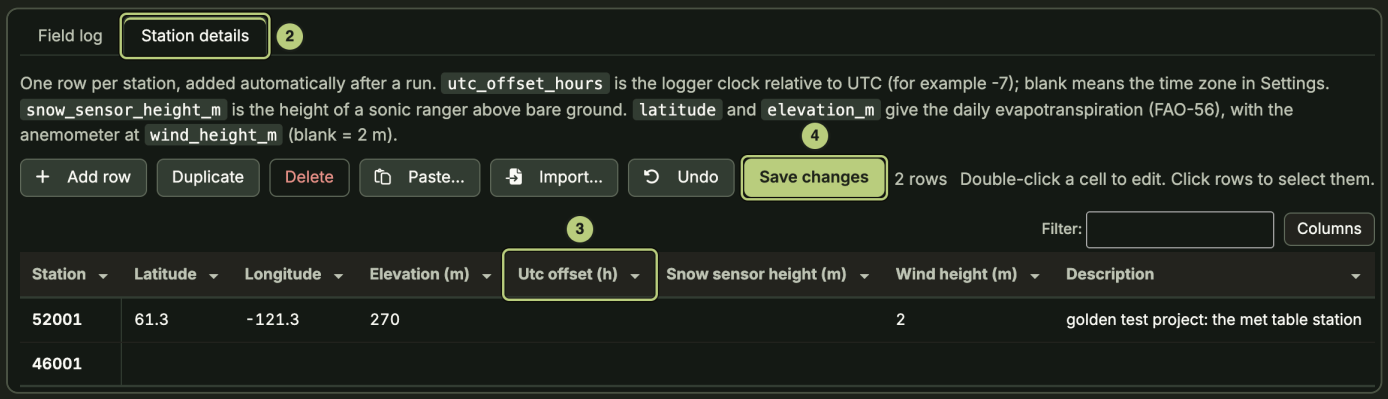
Result: a row in the clock corrections, applied before stitching.

Check: Corrections already set (4) lists it; the finding is gone from **Clocks**; output/audit/timestamp_checks.csv.

8. Set the time zone

When: before the first run, or when a logger kept a different clock.



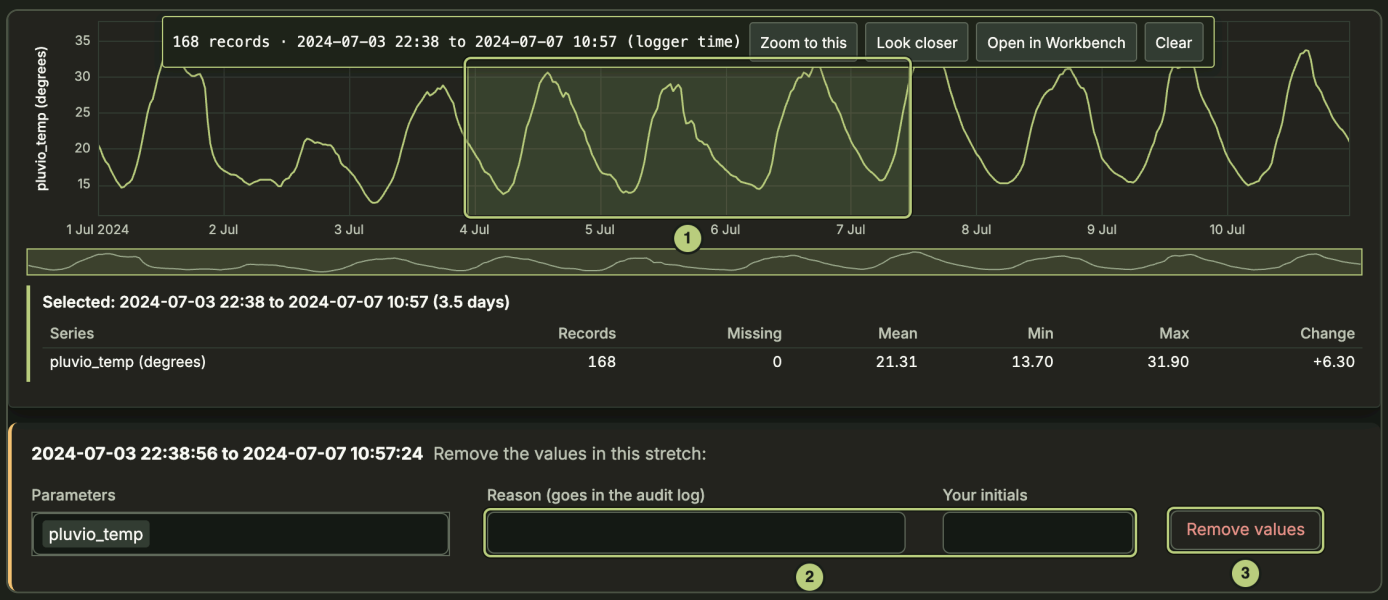


1. **Settings** (gear, top right): **Time zone** as a UTC offset in hours (1), for example -7.
2. A logger on another clock: **Sites and field log** (More pages), tab **Station details** (2). Set its **Utc offset (h)** (3); blank uses the Settings time zone.
3. **Save changes** (4), then **Run**.

Result: every time stamp is moved to the project time zone; exports say which.

9. Remove a bad stretch

When: a sensor was down, being serviced or plainly wrong.



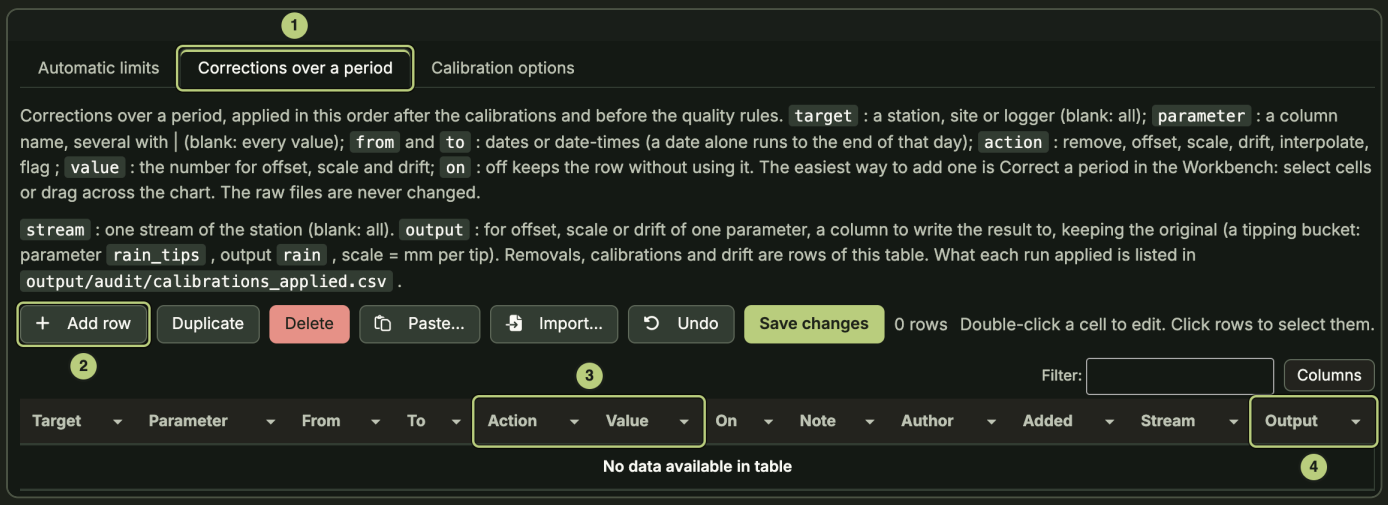
1. **Plot and explore** (More pages): Shift + drag across the stretch (1), then **Open in Workbench**: removing values is done there.
2. In the Workbench the stretch stays selected: **Remove…**, give a reason and your initials, then **Write rows**. To keep the values but mark them, use **Correct a period…** with action flag.
3. Press **Run**.

Result: a row in **Corrections over a period**. The values are blanked on the next run; the old values stay in the audit trail. **Restore…** in the Workbench deletes the row.

Check: the stretch shows as removed on the plot; the **Audit trail** lists the values.

10. Add a calibration

When: a sensor needs a multiplier or an offset over a period, or a tipping bucket needs mm per tip.



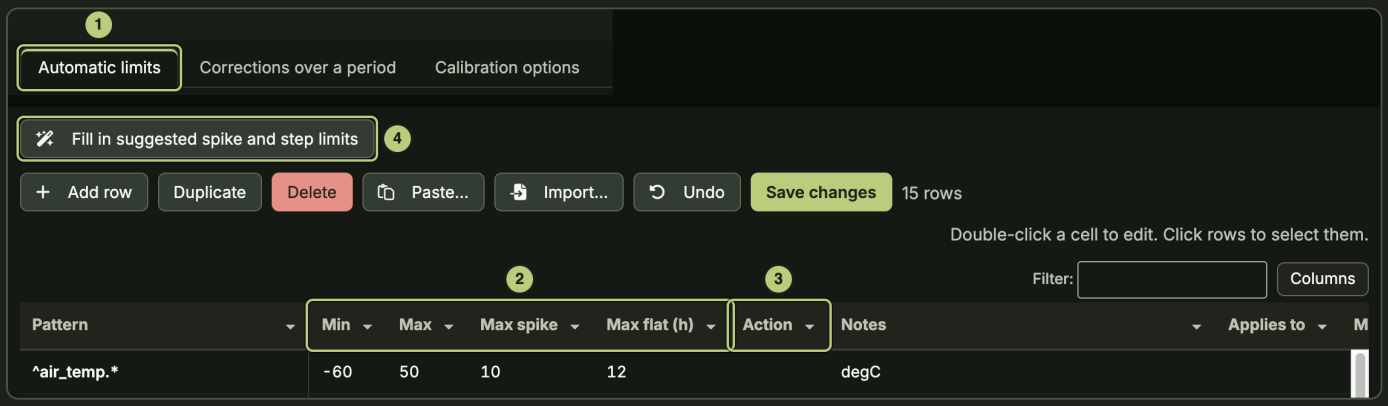
1. In the **Workbench** (calibrations change data values, so the table is edited there): **Calibrations and limits**, tab **Corrections over a period** (1): **Add row** (2).
2. Fill in Target (station), Parameter, From, To, **Action** scale or offset and its **Value** (3).
3. To keep the original, name an **Output** column (4). Tipping bucket: parameter rain_tips, output rain, scale = mm per tip.
4. **Save changes**, then **Run**.

Result: scale multiplies the values by Value, offset adds Value, over the period.

Check: output/audit/calibrations_applied.csv.

11. Change a quality limit

When: a limit removes good values or lets bad ones through.



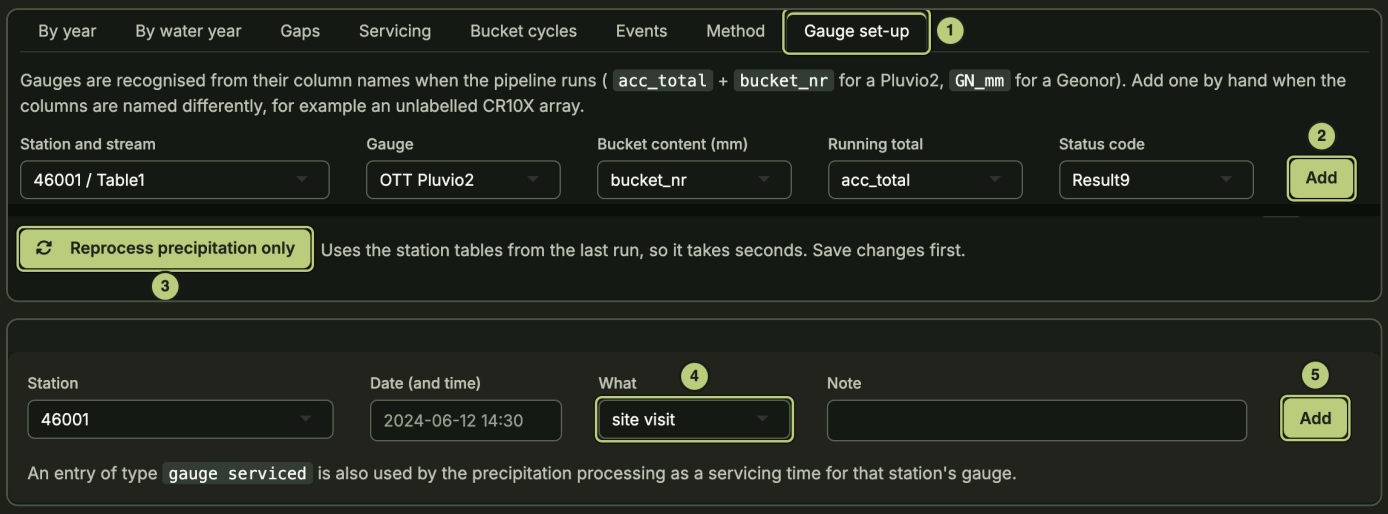
1. In the **Workbench**: **Calibrations and limits**, tab **Automatic limits** (1). **Pattern** picks the parameters, **Applies to** the stations. The Standard view shows the limits.
2. Edit **Min**, **Max**, **Max spike**, **Max flat (h)** or **Max step per hour** (2), and the **Action** (3): blank, remove or flag.
3. **Fill in suggested spike and step limits** (4) fills only blank spike and step cells.
4. **Save changes**, then **Run**.

Result: values outside the limits are removed or flagged and written to the audit trail with the rule. Max step per hour only flags.

Check: **Set-up check** warns of a limit that would remove everything.

12. Set up a weighing gauge

When: a Pluvio2 or Geonor gauge, especially one whose columns are not named acc_total and bucket_nr, or GN_mm.



1. **Precipitation corrections**, tab **Gauge set-up** (1): check the gauge is listed, or set its station, gauge and columns and **Add** (2).
2. **Sites and field log** (More pages), tab **Field log**: for each emptying, top-up or service, pick **What** gauge serviced (4) and **Add** (5).
3. **Reprocess precipitation only** (3).

Result: bucket content turned into precipitation per interval and per day, with undercatch correction where a wind station is set.

Check: tabs **Bucket cycles**, **Servicing** and **Gaps**; days marked low_confidence are read from the bucket.

13. Get evapotranspiration and albedo

When: daily ET (FAO-56 and Priestley-Taylor) or daylight albedo is blank.

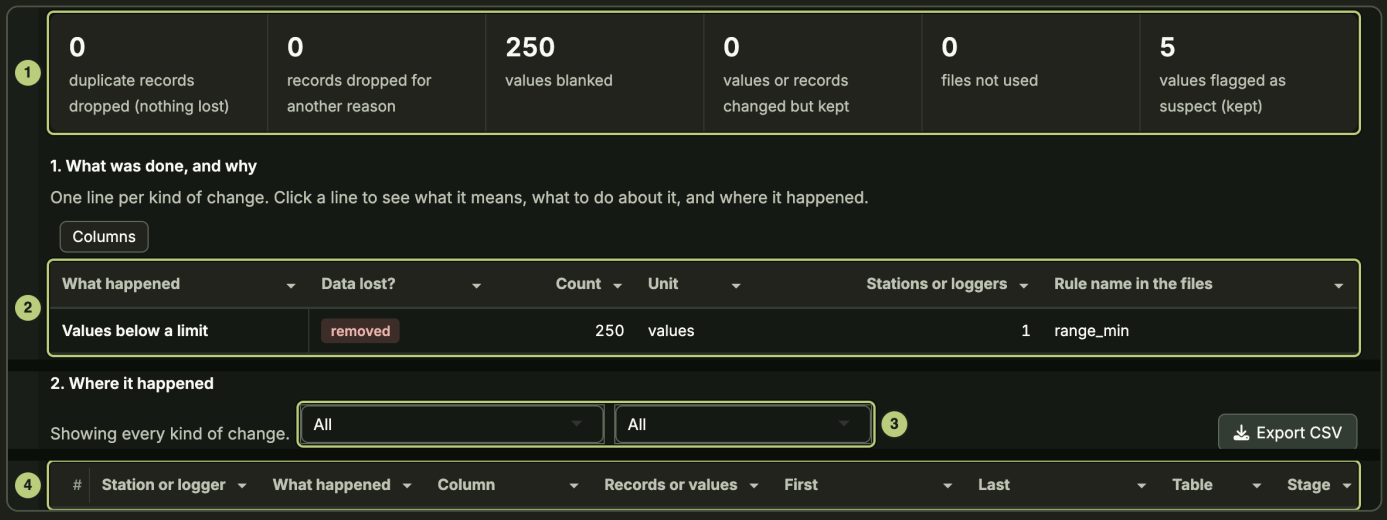


1. **Sites and field log** (More pages), tab **Station details** (1): enter **Latitude**, **Longitude** and **Elevation (m)** (2); **Wind height (m)** (3) if the anemometer is not at 2 m.
2. **Save changes** (4), then **Run**.
3. The daylight limit is **albedo_max_zenith** (6) under **Calibration options** (5): default 70, the sun more than 20 degrees above the horizon.

Result: daily ET from radiation, temperature, humidity and wind; daily albedo from daylight records only.

Check: the daily table in the export. Without coordinates, albedo falls back to incoming shortwave above 50 W/m2; the run log says which was used.

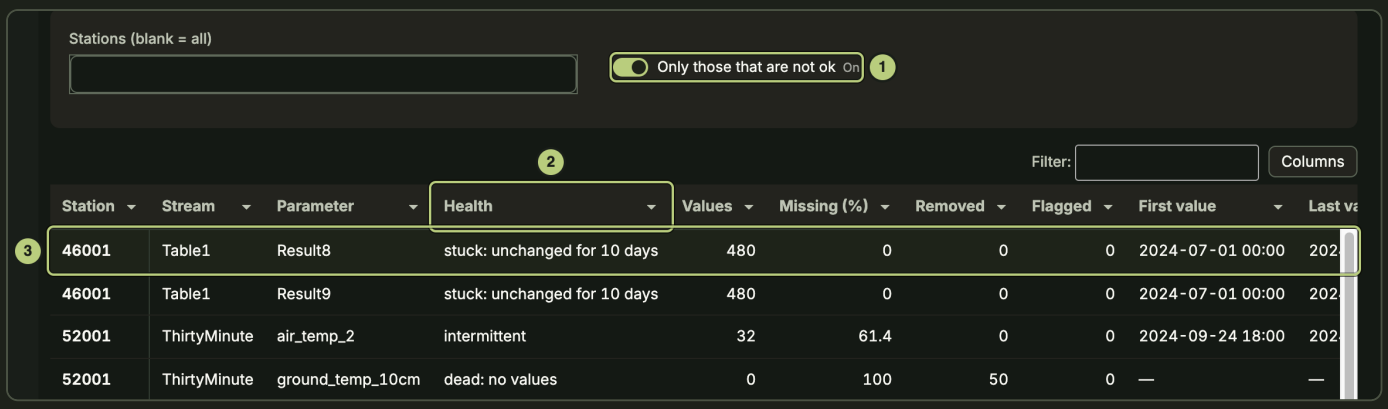
14. Find out why a value was removed



1. **Audit trail** (More pages): read it top down: the totals (1), what was done and why (2), where it happened (3), then the single records and values (4).
2. Click a line in (2) to see what it means, what to do about it and where it happened.

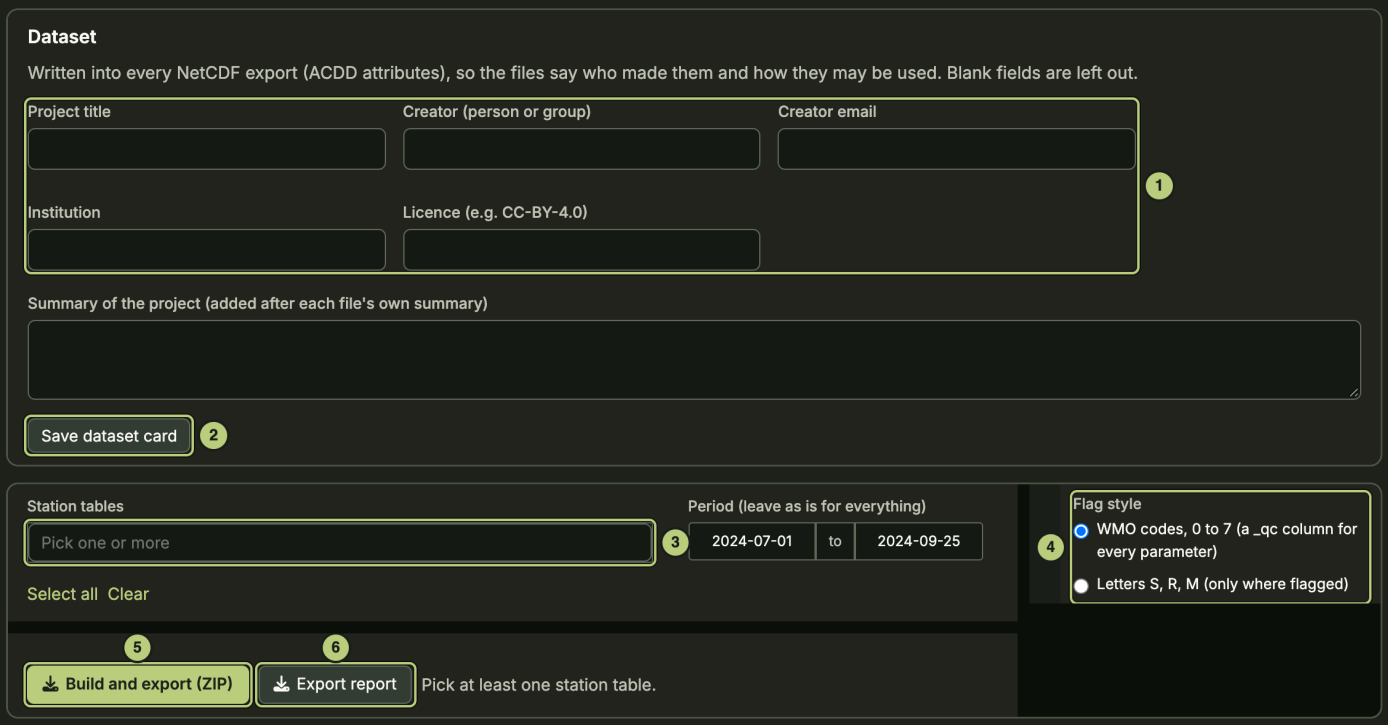
Result: each removed or changed value with its old value, the rule and the set-up row behind it.

15. Find failing sensors



1. **Sensor health** (More pages): with **Only those that are not ok** (1), every parameter that needs a look, worst first. **Health** (2): dead, stopped, failing, dying, stuck, intermittent or suspect.
2. Click a row (3) to open its plot in **Plot and explore**. Remove its bad stretch (task 9) or note it in the field log.

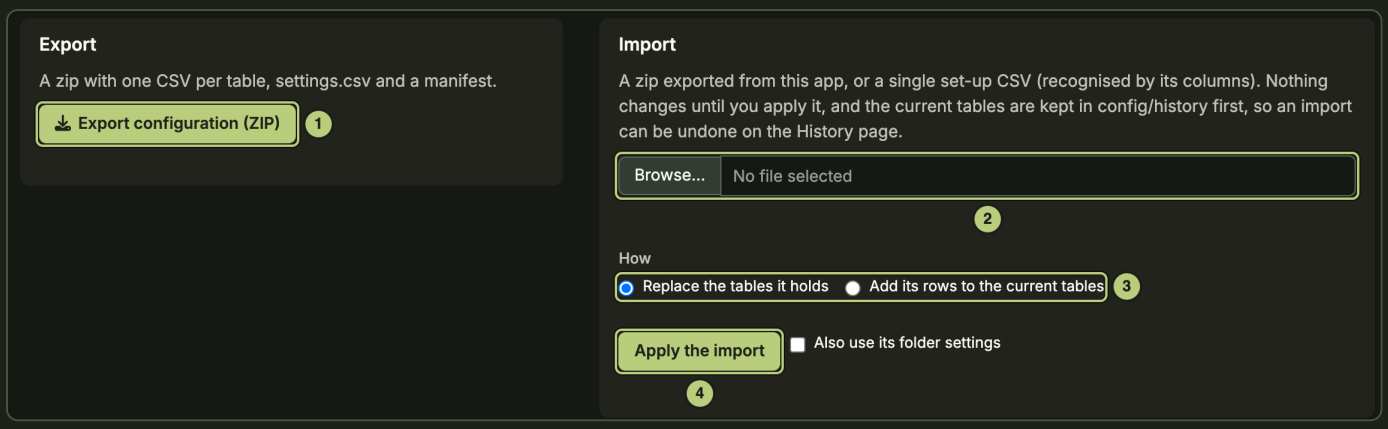
16. Export data and reports



1. **Settings:** fill in the **Dataset** card (1) and **Save dataset card** (2). It is written into NetCDF exports.
2. **7 Report:** pick the **Station tables** (3) and the **Flag style** (4).
3. **Build and export (ZIP)** (5), or **Export report** (6) for the report alone.

Result: one zip that can be used without the app: a tidy CSV per station table with flag columns, daily and monthly summaries, a report per table (HTML, PDF or both), the processing level of each file and a readme.

17. Back up or share the set-up



1. **Configuration** (More pages): **Export configuration (ZIP)** (1) writes every set-up table and the folder settings.
2. In another project: **Browse…** for the zip (2), pick **How** (3), then **Apply the import** (4).

Result: the replaced tables are kept in config/history; the History page undoes the import.

18. Report a problem

1. A notice with a reference (E- and six characters) means part of a page failed; the app carries on.
2. The lifebuoy button (top right) builds a troubleshooting zip: versions, logs and set-up, no raw data. Send it with the reference.